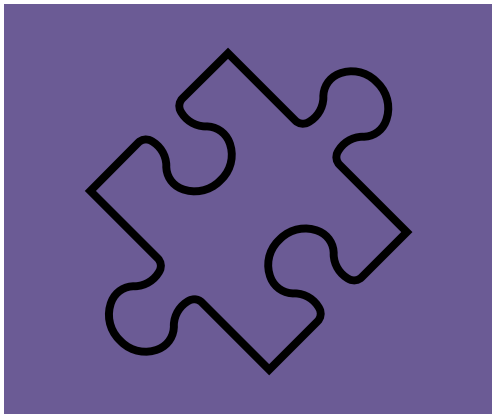
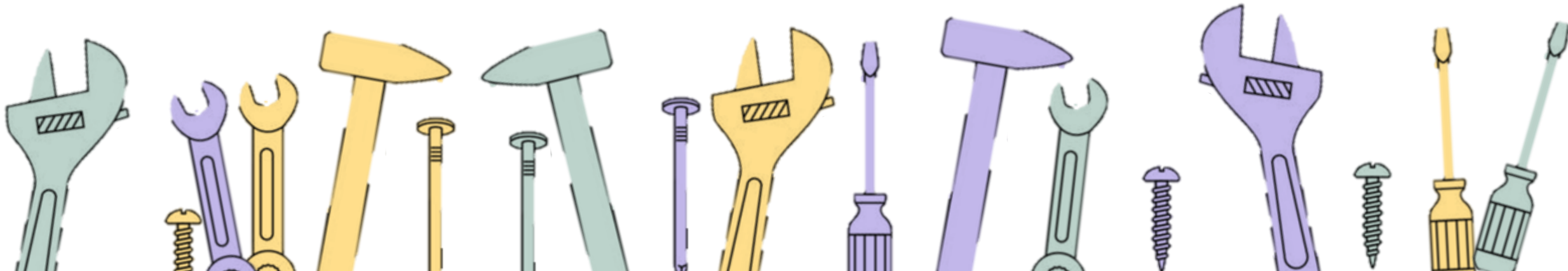


# Horses for courses – The Game



Statistics is not about memorising tests.  
It's about recognising structure.



# Horses for courses – The Game

There are

- 12 Case cards
- 12 Tool cards
- 12 Visualisation cards
- 4 Groups

Each group receives:

- 3 Case cards
- 3 Tool cards
- 3 Visualisation cards

Your group's mission is to:

- 1) Place your **case cards** on the correct board (Inference or Probability)
- 2) Identify the correct **data structure**
- 3) Match the correct **statistical tool**
- 4) Assign the most appropriate **visualisation**

But there's a twist...

You only hold part of the puzzle.

Some of your tools and visualisations belong to other groups' cases — so you'll need to read the boards carefully, challenge decisions, and think like statisticians.

The goal:

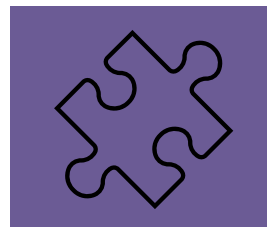
👉 Make sure every case ends up with the correct method and visualisation — and be ready to defend your choices.

Describe the data

Choose the right tool.

Pick the right visualisation.

Defend your reasoning.



# Rules of engagement

-  **It's a game — and games are voluntary.** If you prefer to work alone or prefer not to actively play, that's completely fine. You're welcome to observe, think quietly, or contribute in your own way.
-  **Justify before you stick.** You must explain your reasoning to your group before placing a card.
-  **Read the board.** Some of your tools and visualisations belong to other groups' cases.
-  **Challenge ideas, not people.** If you disagree, explain *why* using the decision tree.
-  **You may challenge one placement from another group.** But you must provide a clear statistical argument.

Are you calculating a probability or analysing collected data?

## PROBABILITY PATH

### 1. Is there a fixed number of repeated trials with two outcomes (success/failure)?

- **Binomial distribution**

- Exact number → Exact binomial probability
- “Or more extreme” → Tail probability

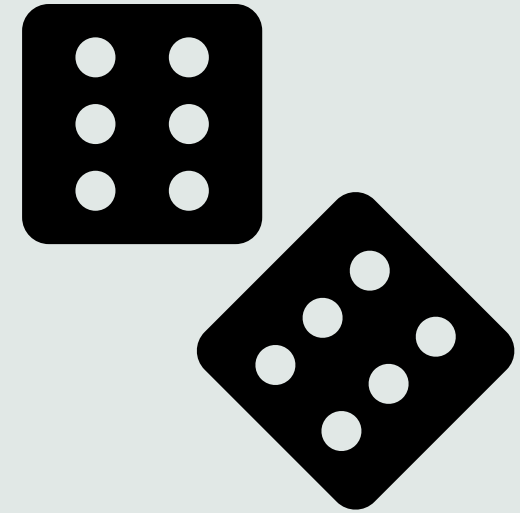
### 2. Are you combining events?

#### “A OR B”

- Are events mutually exclusive?
  - Yes → **Addition rule**
  - No → Addition minus overlap

#### “A AND B”

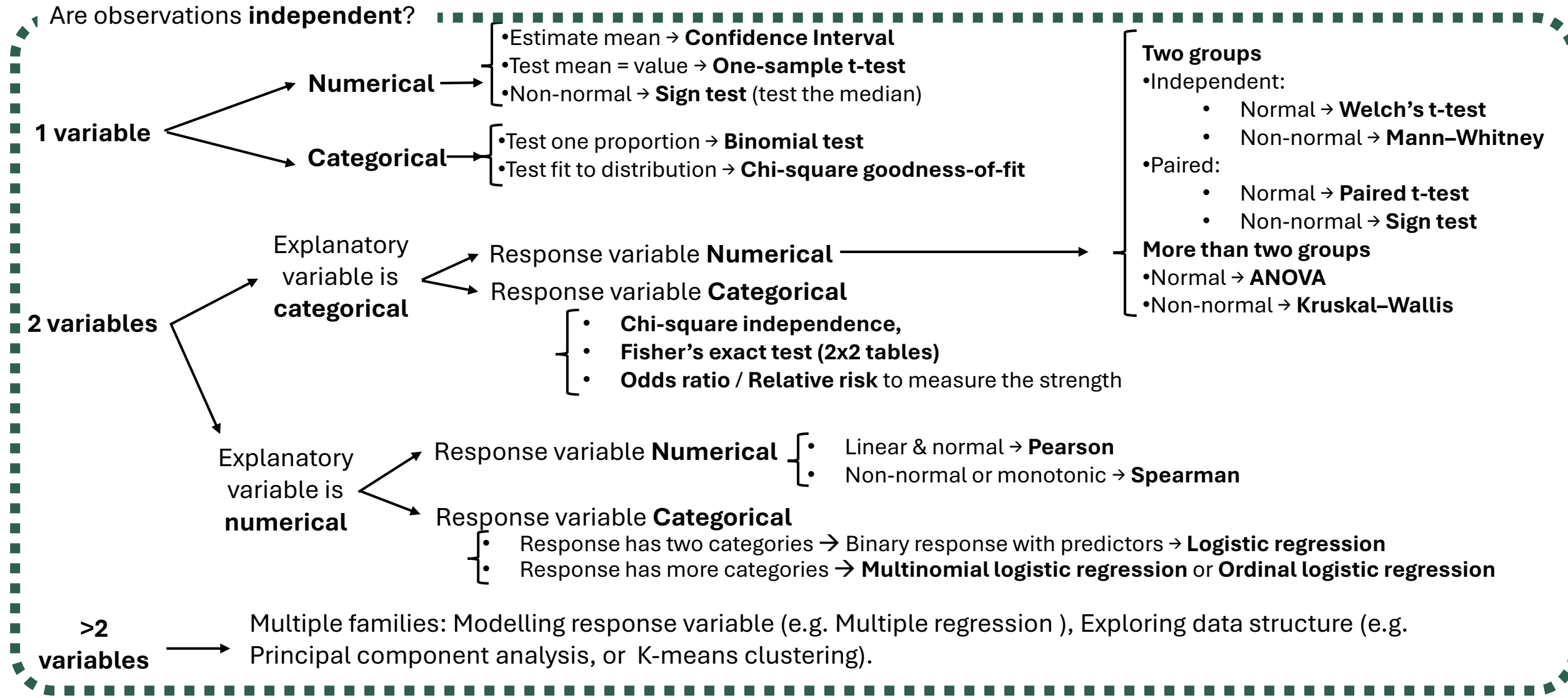
- Are events independent?
  - Yes → **Multiplication rule**
  - No → **Conditional probability**



For analysing collected  
data check next diagram

# HOW TO CHOOSE A MODEL

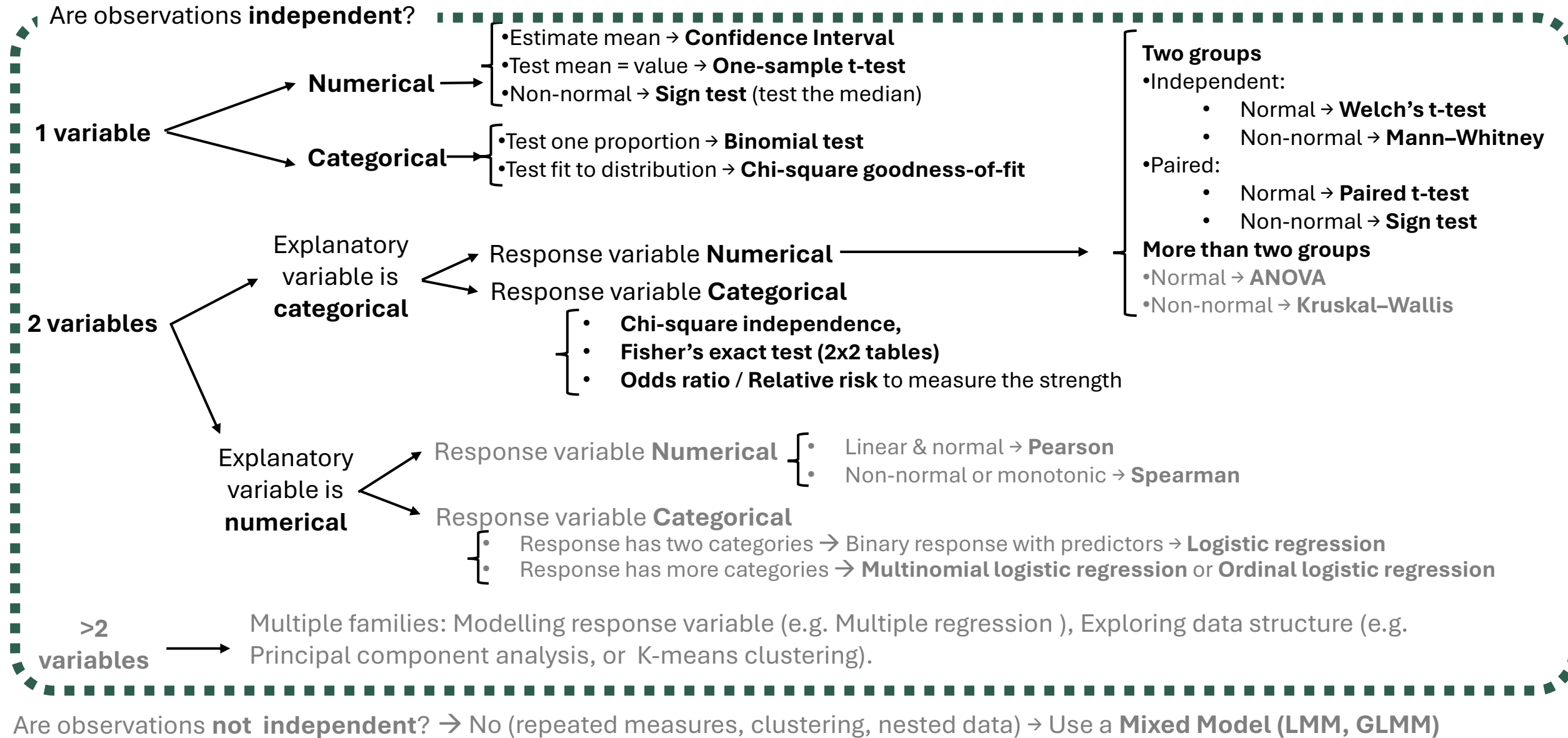
(test-by-test way of thinking about statistics)



Are observations **not independent**? → No (repeated measures, clustering, nested data) → Use a **Mixed Model (LMM, GLMM)**

# HOW TO CHOOSE A MODEL

(test-by-test way of thinking about statistics)



# HOW TO CHOOSE A MODEL

## (modelling version)

If response is:

- Numerical → Linear Model (LM)
- Binary → Binomial GLM (Logistic regression)
- Count → Poisson GLM
- Hierarchical → Mixed Model (LMM / GLMM)

# Meta-learning exercises

Which distinction caused the most confusion today?

- Probability vs inference?
- GOF vs independence?
- Paired vs independent?
- Normal vs non-normal?

**Before choosing a method, always identify:**

- 1.One variable or two?
- 2.Numerical or categorical response?
- 3.Independent or paired?
- 4.Normal or not?
- 5.Probability or inference?

If you can answer those, the method follows.

For you to practice: Check that you understand the difference between:

- Binomial test vs  $\chi^2$  GOF (Case 2 vs Case 3)
- Probability vs p-value thinking (Case 9 vs inference cases)
- Welch vs ANOVA
- Mann–Whitney vs Welch
- Independence vs multiplication rule (Case 7 vs Case 12)

Suggested consolidating exercise: write down your own example cases on the decision tree.